

Module 01

Introduction to AI-Driven Software Quality Assurance (SQA)

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Course: AI Driven SQA
Module: Module 01
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Submission Date: 04 August 2026

Q1. What is Software Quality Assurance (SQA)?

Define SQA Explain why quality is important in software development.

Answer: SQA is a planned, systematic set of activities that ensure software meets defined quality standards.

SQA focuses on ensuring quality throughout the development process and minimizing defects by addressing potential problems early, rather than relying solely on testing the completed software.

Software quality is important because it builds customer trust, protects business revenue, improves user satisfaction, supports security and compliance, and reduces long-term maintenance cost.

Q2. Differentiate between the following:

QA vs QC

Aspect	Quality Assurance (QA)	Quality Control (QC)
Focus	Process-oriented	Product-oriented
Goal	Prevent defects	Identify defects
Approach	Proactive	Reactive
Activities	Audits, standards, process reviews	Inspection, reviews of the product
Responsibility	Entire team	QC / QA team
Example	Defining a code review checklist	Reviewing a build against that checklist

QC vs Testing

Aspect	Quality Control (QC)	Testing
Focus	Product-oriented	Execution-oriented
Goal	Identify defects	Find bugs & verify behavior
Approach	Reactive	Reactive
Activities	Inspection, reviews of the product	Test case execution, exploratory testing
Responsibility	QC / QA team	Testers / Automation Engineers
Example	Reviewing a build against that checklist	Running login test cases on that build

QA vs Testing

Aspect	Quality Assurance (QA)	Testing
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Q3. How is AI changing Software Testing?

AI is changing software testing by making the testing process more intelligent, efficient, and reliable. It helps improve software quality by supporting different testing activities throughout the Software Development Life Cycle (SDLC).

From Scripted to Intelligent: Traditional automation follows fixed scripts. AI-driven testing understands intent, adapts to change, and learns from past runs.

Self-Healing Automation: When a locator breaks due to UI changes, AI automatically finds the new element and updates the test, reducing maintenance.

AI Across the SDLC: AI now assists in requirement analysis, test case generation, Code review, execution, and defect triage—not just execution.

Intelligent Reporting: AI summarizes failures, clusters similar defects, and highlights root causes instead of raw pass/fail logs.

Q4. Explain the responsibilities of:

QA Engineer, Software Test Engineer, AI Quality Engineer. Describe the daily activities of each role.

QA Engineer	
Responsibilities	Daily Activities
➤ Ensure software quality throughout the Software Development Life Cycle (SDLC). ➤ Review requirements for ambiguity, completeness, and testability. ➤ Define quality standards and improve the development process. ➤ Prevent defects before software is released. ➤ Work with developers and project managers to improve software quality.	❖ Review requirements and user stories. ❖ Prepare test plans and test cases. ❖ Review test results. ❖ Monitor software quality. ❖ Participate in sprint planning and meetings. ❖ Support process improvement.

Software Test Engineer	
Responsibilities	Daily Activities
➤ Verify that the software functions according to the specified requirements. ➤ Execute manual and automated test cases. ➤ Find, report, and track software defects. ➤ Perform different types of testing such as functional, regression, integration, and API testing. ➤ Confirm that defects are fixed correctly.	❖ Execute test cases. ❖ Perform functional testing. ❖ Perform regression testing. ❖ Report bugs. ❖ Retest fixed issues. ❖ Update test reports. ❖ Work closely with developers.

AI Quality Engineer	
Responsibilities	Daily Activities
➤ Test AI-powered applications and AI features. ➤ Validate AI-generated output before software release. ➤ Test AI responses for: i. Accuracy ii. Hallucination iii. Context iv. Prompt Understanding v. Security ➤ Ensure AI systems provide correct, safe, and reliable responses.	❖ AI-assisted requirement analysis. ❖ AI-generated test cases. ❖ AI-generated test data. ❖ AI-assisted test execution. ❖ AI-assisted bug reporting. ❖ AI-assisted automation script generation. ❖ AI-assisted documentation. ❖ Review and verify AI-generated output before production deployment.

Q5. Draw the career roadmap of a QA professional.

The career roadmap of a QA professional shows the growth path from an entry-level QA role to leadership positions

Junior QA Engineer



QA Engineer



Automation QA Engineer



Senior QA Engineer /AI-Augmented QA Engineer



AI Quality Engineer



Senior AI Quality Engineer



AI QA Lead / AI Quality Lead



AI Quality Manager



Head of AI Quality

